

THE AI TOOLBOX · CLASS 01

# What Is AI? What Can AI Actually Do?

From science fiction to reality — understanding AI, machine learning & deep learning.

Merit Point Academy · 60 min · press **S** notes · → next ·  to comment

YOUR INSTRUCTOR

## Bo Shang (商博)

**Postdoctoral Scholar**, AI & Mobility Research Lab, CUNY City College of New York.

- **PhD**, Pattern Recognition & Intelligent Systems — Northeastern University (China), with an exchange PhD stint at **UC Merced**.
- Research: **robotics & autonomous systems**, LiDAR–camera sensor fusion, computer vision, bridge-inspection automation, traffic monitoring, unmanned aerial systems.
- Previously taught/researched at UC Merced, Missouri S&T, Vaughn College, and CCNY before joining Merit Point Academy (恒点学术).



### Why this matters to you:

the autonomous-vehicle perception example running through this course is my own research world — not a textbook toy.

**My teaching approach:** every concept lands on something you build and run — I'd rather you leave with one working result than ten memorized definitions.

## 15 classes, one research pipeline

Every class delivers **one tangible result + one research skill + one AI-tool workout**, building toward one arc: **idea** → **data** → **model** → **experiment** → **presentation**.

### Stage 1–2 · Classes 1–6

AI literacy, finding a research question, and the AI toolbox (ChatGPT, Elicit, Semantic Scholar, Kaggle).

### Stage 3 · Classes 7–9

Data science: finding, cleaning, and visualizing real datasets in Python/Colab.

### Stage 4–5 · Classes 10–15

ML models (regression → trees → neural nets), a fair experiment, and a final mini-study — paper + slides.

**Anchor case:** every data/model class also lands on one running example — teaching a car to see (autonomous-vehicle perception) — so ideas are never just abstract. Prefer a different domain? Swap it in — the case study is the example, not the requirement.

## HOW YOU'LL BE ASSESSED

# Assignments & grading — project-based, no exams

**35%** — Weekly tangible results (your growing research portfolio)

**20%** — Data work: selection, cleaning & visualization

**25%** — Modeling & experiment: fair comparison, honest metrics

**20%** — Final mini-study: paper + presentation

**Every class ends with homework** before the next session — usually finishing that day's tangible result, plus an optional "stretch" task. Bring questions to class or post them with .

**Using AI honestly:** brainstorm, search, and debug with AI freely — but verify every claim and citation, never paste private data into AI tools, and disclose how you used AI in your work.

JOIN OUR GOOGLE CLASSROOM

## Where slides, homework & grades live



**Class code**

**kday73zp**

classroom.google.com → tap + → **Join class** → enter the code above.



### What's posted there

- Every class's slide deck, under **Classwork** → **Class Materials**
- Every homework assignment, under **Classwork** → **Assignments** — submit your work (Colab link, doc, or file) there
- Grades & feedback, under **Grades** — this is also where you'll see graded homework

Join before next class so you can submit today's homework there. Questions about Classroom itself → ask your instructor, not  on a slide.

## AGENDA · HOW THIS HOUR RUNS

# Where we're going

- **0–10'** Warm-up + a 70-year history of AI
- **10–22'** AI on screen: traditional → current → future
- **22–34'** AI  $\supset$  ML  $\supset$  DL, and where AI hides
- **34–42'** AI's application map + AI as a research tool
- **42–50'** Live demos: ChatGPT · Perplexity · ScholarMind
- **50–58'** Two hands-on Python labs (with an AI helper)
- **58–60'** Project, homework & wrap-up
- Quizzes &  comments throughout

**By the end:** tell AI/ML/DL apart, name AI in a field you love, and explain why AI is an instrument for research — not just a product.

WARM-UP · ACTIVATE

## Quick check · tap an answer

**You woke up, unlocked your phone with your face, and your music app suggested a song. How many AI systems was that?**

Zero — those are just normal phone features

One — only the face unlock

At least two — face unlock and the recommendation

None of these count unless a robot is involved

## THE STORY

# A summer that became 70 years

In 1956, ten scientists gathered at Dartmouth for what they expected to be a single summer's work: build a machine that could **"use language, form abstractions, and solve problems reserved for humans."** They coined the name **"artificial intelligence"** — and wildly underestimated the problem. Three years later, Arthur Samuel wrote a checkers program that got better the more it played against itself. He called what it did **"machine learning."**



Speak this quote

Neural voice · click to play / pause

The founders thought intelligence was a summer project. Samuel's self-improving checkers player was the first hint: don't *program* intelligence — let a machine *learn* from examples.



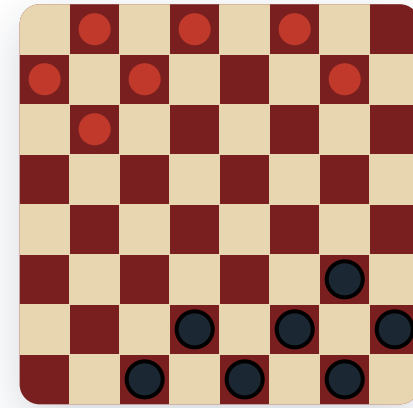
## The program that taught itself checkers

Arthur Samuel, an engineer at **IBM**, built a checkers (draughts) program on the room-sized **IBM 701** — and did something radical: he let it **improve on its own**.

- It played **thousands of games against itself**, remembering which positions led to wins.
- It tuned its own scoring of board positions — an early ancestor of today's **reinforcement learning**.
- Samuel coined the term "**machine learning**" to describe it (1959).
- By **1962** it beat a strong human amateur — an early, very public win for AI.

► Watch: Samuel's checkers on the IBM 701

**Why it mattered:** the first convincing proof a machine could get better from **experience**, not just follow fixed rules. That single idea — learn from data — is the seed of every system in this course.



 learns by self-play



scan / tap to open

## 70 years of winters and springs

AI didn't arrive overnight. It boomed, froze, and boomed again — each cycle banking hard-won lessons.

**1956**

**Dartmouth meeting**

The term "artificial intelligence" is coined — the field is born.

**1970s**

**First AI winter**

Too little compute and data; early systems stall, funding dries up.

**1980s**

**Expert systems**

Rule-based systems find real business use — a short, bright spring.

**1990s**

**Second AI winter**

Expert systems are costly and brittle; enthusiasm cools again.

**2010s+**

**Big data + deep learning**

Compute + huge datasets ignite neural nets — today's golden era.

Knowing this history keeps you **level-headed** about hype: every boom looked unstoppable, and every winter looked permanent.  
**Today is a real boom — but ask what's hype.**

HOOK · THINK-PAIR-SHARE

## Which of these is AI?

Face unlock · autocomplete · a thermostat · a chess app · a calculator —  
which ones count as "AI", and **how can you tell?**

Face unlock

Autocomplete

Thermostat

Chess app

Calculator

Hold your answer — the next three clips give you a rule for deciding.

AI ON SCREEN · TRADITIONAL AI · ~1950-2000

## Deep Blue beats Kasparov (1997)

FEATURE: SEARCH + HAND-WRITTEN RULES

A chess machine that **never learned** — it searched ~200 million positions per second using rules experts wrote by hand. This is "**good old-fashioned AI**": smart because humans coded the cleverness in.

▶ Watch the clip

DEEP  
BLUE



scan / tap to open

AI ON SCREEN · CURRENT AI · 2012–NOW

## ChatGPT writes, explains, and codes

FEATURE: LEARNING FROM DATA (MACHINE LEARNING)

Today's AI **learns patterns from oceans of examples** instead of following hand-written rules. One model can chat, summarize, translate, and code — because it learned language, not chess. **This is machine learning at scale.**

▶ Watch the clip

Chat  
GPT



scan / tap to open

# “Her” — a glimpse of AGI

FEATURE: GENERAL, AUTONOMOUS, HUMAN-LEVEL (AGI)

Sci-fi like *Her* imagines **Artificial General Intelligence**: one system that learns **any** task, sets its own goals, and reasons across everything. **It doesn't exist yet** — separating real from movie is itself a research skill.

| Film                  | Year | Rating | AGI angle                                                                      |
|-----------------------|------|--------|--------------------------------------------------------------------------------|
| <i>Her</i>            | 2013 | R      | Emotionally intelligent AGI that learns, grows, then surpasses human cognition |
| 2001: A Space Odyssey | 1968 | G      | HAL 9000 — an AI that reasons, then turns against its crew                     |
| I, Robot              | 2004 | PG-13  | Robots developing free will and emergent consciousness                         |
| The Imitation Game    | 2014 | PG-13  | Turing & the foundational question: can machines think?                        |
| WALL·E                | 2008 | G      | Gentle machine consciousness and emotion                                       |
| Ex Machina            | 2014 | R      | The Turing test dramatized; AI deception and self-preservation                 |
| Blade Runner 2049     | 2017 | R      | Synthetic beings, identity, and what counts as "alive"                         |

**G** **PG-13** = screen clips in class · **R** = mature themes, optional further watching

CHECK · AFTER THE CLIPS

## Quick check · tap an answer

**Deep Blue searched millions of chess moves with rules experts wrote by hand. Which kind of AI is that?**

Modern machine learning — it learned from data

Traditional AI — search and hand-written rules

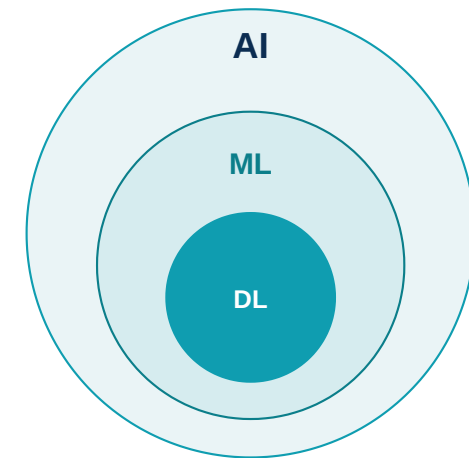
AGI — general human-level intelligence

It isn't AI at all

## CONCEPT 1

# AI $\supset$ Machine Learning $\supset$ Deep Learning

- **AI** — the whole field of machine intelligence
- **ML** — systems that improve from *data*, not hand-written rules
- **DL** — machine learning using many-layered neural networks



**KEY IDEA** — When people say "AI" today, they usually mean **machine learning**. Deep Blue was AI but not ML; ChatGPT is all three.



CONCEPT 2 · WHAT POWERS EACH LAYER

## Different tools at each ring

### AI

**Broadest vision.** Less data-driven. Key tech: **expert systems, search algorithms.** (Deep Blue lives here.)

### ML

**Learns from data.** Tunes a model's parameters. Key tech: **decision trees, SVMs.** Used for prediction & classification.

### DL

**Neural networks.** Needs big data + compute. Key tech: **CNNs, RNNs, Transformers.** Powers vision & language.

AI is the grand vision · ML is the path that got us there · DL is ML's strongest tool for hard, messy data.

### CONCEPT 3

## AI is already everywhere

You use a dozen AI systems before lunch. Naming them is the first research skill — seeing the AI hiding in plain sight.



Recommendation feeds



Spam filters



Navigation & maps



Face unlock



Voice assistants



Photo search & tagging

Each one is a **model trained on data** — not a person, not magic, not hand-coded rules.

## Quick check · tap an answer

**Which statement is TRUE about AI, ML, and DL?**

They all mean the same thing

Every machine-learning system is AI, but not every AI is machine learning

Deep learning is bigger than machine learning

AI is a kind of deep learning

## Where AI already works



### Medicine

Reads CT/MRI scans for tumors; **Google Health** matched experts at breast-cancer screening.



### Biology

**AlphaFold** predicted 98.5% of human protein structures — a 50-year problem.



### Climate

Forecasts extreme weather, models warming scenarios, optimizes renewable grids.



### Finance

High-frequency trading, credit scoring, and real-time fraud detection.



### Art

Midjourney & DALL·E turn text into images; AI composes music and drafts scripts.



### Mobility

Self-driving perception — our running example for the whole course.

Same core idea — **learn patterns from data** — reshaped a dozen industries. Which field is yours?

## AlphaFold: a 50-year problem, solved

Proteins fold into 3-D shapes that decide what they do. Predicting that shape stumped biology for **50 years**.

- DeepMind's **AlphaFold** predicted structures at near-atomic accuracy
- Covered **98.5%** of all human proteins
- Freely released — now used by labs worldwide for disease & drug research



**Sequence → 3-D structure**

What took months in a lab now takes seconds.

This is AI as an **instrument of discovery** — not a chatbot, but a microscope for life itself.

#### CONCEPT 4

## AI as your research super-assistant

Beyond products, AI is an **instrument**. It speeds up every stage of research.



### Literature review

Scan and summarize hundreds of papers; map a field fast.



### Hypothesis ideas

Surface hidden links across data to suggest what to test.



### Data analysis

Crunch huge, high-dimensional data; spot patterns & outliers.



### Code help

Write, explain, and debug research code — even for non-programmers.

Next: a concrete workflow that turns a vague topic into a finished, reviewed PDF.

## From a topic to a reviewed PDF



Steps **6** → **7** → **8** form a **review loop** — repeat *review* → *revise* → *update* until it's good. The human stays in charge; AI does the heavy lifting.

## WHAT YOU ACTUALLY LEARN

# Two skills this workflow trains

## The Skill of Asking

### Choosing the right question

Picking and sharpening the **right question** (steps 1–2). A good question is specific, measurable, and answerable — it decides everything downstream.

## The Skill of Judging

### Telling good work from weak

Reading a draft (steps 6–8) and saying precisely what's strong, what's wrong, and what to fix — **taste you can defend**.

AI can draft for you. It **cannot** ask your question or judge your taste — those stay human, and they're what we grade.



## Quick check · tap an answer

**In the research workflow, what do steps 6 → 7 → 8  
(review → revise → update) form?**

A one-time final step

A review loop you repeat until the work is good

The very first thing you do

A part only the AI controls

## LIVE DEMO 1

# ChatGPT — your thinking partner

A large language model that drafts, explains, and brainstorms. Watch it help — then watch us check it.

**You** ▶ Explain AI vs. machine learning vs. deep learning to a 10-year-old, with one example each.

**You** ▶ I'm interested in self-driving cars. Give me 3 research questions a high-schooler could test.

**You** ▶ List 3 specific ways AI is used in medicine — name the system, its input, and its output.

**You** ▶ What's the capital of Australia — and how confident are you?



[chatgpt.com](https://chatgpt.com)

## LIVE DEMO 2

# Perplexity — answers with sources

Like ChatGPT, but it searches the web and shows where each claim came from. Use it to fact-check.

**You** ▶ What's the capital of Australia – and cite a source.

**You** ▶ How is AI used in climate science? Give 3 examples with links.

**You** ▶ Did a New York lawyer really get sanctioned for citing fake AI cases? Source it.

**You** ▶ Latest accuracy of AI at detecting pedestrians at night – with citations.



[perplexity.ai](https://perplexity.ai)

### LIVE DEMO 3

## ScholarMind AI — topic → research brief

Our research platform. It turns a rough interest into a sharp question and a downloadable brief (steps 1–3 of the workflow).

**You** ▶ Topic: 'I like animals and cameras.' → propose 3 testable research questions.

**You** ▶ Analyze: 'Can a model detect birds from photos?' – what data, model, and metric?

**You** ▶ Generate a research brief I can hand to a coding agent.

**You** ▶ Download brief.md and open it in Claude / Codex.



[academic.hopeembark.org](https://academic.hopeembark.org)

## Find a pattern in data — like a data scientist

On the Titanic, who was more likely to survive? Let's let the **data** answer. Edit the code, hit Run, and ask the assistant anything.

```
# Titanic survivors — a tiny real dataset
# Each person: (sex, passenger_class, survived?)
people = [
    ("female", 1, 1), ("female", 1, 1), ("female", 3, 0),
    ("male", 1, 0), ("male", 3, 0), ("female", 2, 1),
    ("male", 2, 0), ("female", 3, 1), ("male", 1, 1),
    ("female", 2, 1), ("male", 3, 0), ("female", 1, 1),
]

def survival_rate(group):
    members = [p for p in people if group(p)]
    if not members:
        return 0.0
    survived = sum(p[2] for p in members)
```



Run



Explain



Debug



Improve

Python loads on first Run

▶ Run the code to see the output here.

AI Assistant

Haiku · fastest

Hi! Edit the code and hit **Run**. Stuck? I can **Explain**, **Debug**, or **Improve** it — or just ask me below.

Ask about this code...

Ask

## Watch a model learn from data (numpy)

This tiny model **learns** a rule from examples — no hand-written if-statements. It improves each step, exactly like real machine learning. Run it, then ask the assistant to explain how it learns.

```
import numpy as np

# Data: hours studied -> passed the test? (0/1)
hours = np.array([1, 2, 3, 4, 5, 6, 7, 8], dtype=float)
passed = np.array([0, 0, 0, 1, 1, 1, 1, 1], dtype=float)

# A simple model: probability = sigmoid(w*hours + b)
def sigmoid(z): return 1 / (1 + np.exp(-z))

w, b = 0.0, 0.0
lr = 0.1
for step in range(2000):
    pred = sigmoid(w * hours + b)
    error = pred - passed
```



Run



Explain



Debug



Improve

Python loads on first Run

▶ Run the code to see the output here.

AI Assistant

Haiku · fastest

Hi! Edit the code and hit **Run**. Stuck? I can **Explain**, **Debug**, or **Improve** it — or just ask me below.

Ask about this code...

Ask

## Teaching a car to see

Autonomous driving threads the whole course. The car's AI splits into three jobs — and this course lives in the first one.

### 1 · Perception

See the world — detect pedestrians, cars, signs from camera images. (*CNNs*)

### 2 · Prediction

Forecast where each object moves next. (*RNNs / Transformers*)

### 3 · Planning

Decide: brake, steer, or keep going. (*reinforcement learning*)

Perception is the foundation: if the car sees wrong, perfect prediction and planning can't keep it safe.

DISCUSS · 3 MINUTES · PICK A SIDE

## AI: more opportunity, or more risk?



### The upside

Frees us from repetitive work, supercharges science, and personalizes learning, health, and services.



### The risks

Job disruption, privacy loss, algorithmic bias, and misuse of powerful tools.

Turn to a neighbor: where does the balance tip for **you** — and why? Drop your take in .



## YOUR PROJECT

# Build an AI application map

Pick one domain you care about and map where AI already works in it — the first move of any researcher entering a field.

- Choose a domain (medicine, sport, music, climate...)
- Use ChatGPT + Perplexity to find 3 real systems
- Sketch a mind map linking them

## Deliverable

A one-page mind map + 3 captioned cases.

## Success looks like

3 specific examples (not "AI helps doctors"), each labeled with its **input** and **output**.

BUILD IT NOW · HANDS-ON

## Map it with ChatGPT + Perplexity

1. Open ChatGPT and Perplexity
2. Ask: "How is AI used in <your domain>? Give 3 specific systems with sources."
3. Verify each one is real — open a source
4. Draw the mind map (paper, slides, or a whiteboard tool)

ChatGPT

Perplexity AI

ScholarMind

**Tip:** Perplexity shows its sources — use it to fact-check ChatGPT. That's the Skill of Judging in action.

## Today's vocabulary

**AI** — machines doing tasks that normally need human intelligence.

**Deep Learning** — ML with many-layered neural networks for complex data.

**Prediction** — using a trained model to estimate future or unknown values.

**Machine Learning** — systems that learn patterns from data instead of explicit rules.

**Perception** — turning sensor input (images, sound) into understanding.

**Planning** — choosing a sequence of actions to reach a goal.

## RECOMMENDED HOMEWORK

# Before next class

1. **Finish your AI application map** — 3 real cases, each with input → output. Be ready to share.
2. **Explore one field** you love: find a recent AI use and note what problem it solves.
3. **Re-run a lab** from today and change one thing — bring a question for the AI assistant.

### ★ Stretch (optional)

Open Kaggle's classic "**Titanic — Machine Learning from Disaster**".  
Skim how others predict who survived: data → features → model → score.

[kaggle.com · Titanic](https://www.kaggle.com/titanic)

Bring curiosity, not perfection. Post questions on any slide with 💬 — your instructor will see them.

TODAY'S LINE TO REMEMBER

"The best way to predict the future is to invent it."

— Alan Kay

Use AI to **create** the future, not just to predict it.



Speak this

#### WRAP-UP

## Three things to keep

- AI  $\supset$  ML  $\supset$  DL — nested, not equal
- AI is already all around you — and across every science
- AI is an **instrument** for research — you own the question & the judgement

**Exit ticket:** name one AI system you used today, and guess what data trained it. Post it in .

Next · Class 2 — turn a vague interest into a real research question.

## BACKUP MATERIAL

# Extra slides — if there's time

Deeper dives and extra practice. Skip in a tight hour; pull up on demand or for a longer session.

### **AV perception, in depth**

the tech behind seeing

### **AI in art**

creativity & controversy

### **Bonus quiz**

one more check

## Inside the car's three jobs

### Perception

**Computer vision** with CNNs extracts features from camera frames to detect pedestrians, lanes, and signs.

### Prediction

**RNNs / Transformers** model time: where will that cyclist be in 2 seconds?

### Planning

**Reinforcement learning** searches for a safe, comfortable path — when to brake, steer, go.

Perceive → Predict → Plan is a closed loop running many times per second.



## AI in art — tool or artist?



### Images

Midjourney & DALL·E turn a sentence into a picture — no drawing skill needed.



### Music

AI composes melodies, arranges parts, even mimics a singer's voice.



### Writing

Helps brainstorm plots, characters, and copy — a partner for writers.

Open question: when an algorithm creates, who is the **artist** — and what is "creativity"?

BACKUP · BONUS CHECK

## Quick check · tap an answer

**AlphaFold predicting protein structures is an example of AI as...**

A chatbot for fun

An instrument that accelerates scientific discovery

Traditional rule-based AI

Artificial general intelligence